FluidAgent Pro

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Abstract

This paper introduces FluidAgent Pro, a computational fluid dynamics (CFD) solver developed for simulating complex flow phenomena. The solver's core logic is implemented in C++, with post-processing and analysis conducted using Python. To assess its accuracy and reliability, the solver was validated against a standard benchmark case. The numerical results obtained from the simulation show a good agreement with the established benchmark data, demonstrating the solver's capability for predictive simulations. This work establishes a foundation for further development and application of FluidAgent Pro to more complex engineering and scientific problems.

1 Introduction

Computational Fluid Dynamics (CFD) has become an indispensable tool in science and engineering for the analysis and design of systems involving fluid flow. Particularly in the study of multiphase flows, such as gas-solid systems, CFD provides insights that are often difficult or expensive to obtain through experimental methods alone. Accurate prediction of these flows relies heavily on the underlying physical models and the robustness of the numerical solver.

The primary objective of this study is the development and verification of a new CFD solver, named FluidAgent Pro. The development process emphasized a robust C++ implementation for the core computational engine to ensure high performance and scalability. The verification of the solver is a critical step to ensure that the implemented models and numerical schemes are correct. In this work, we present a validation study by comparing the simulation results from FluidAgent Pro with widely accepted benchmark data from the literature. This comparison serves to quantify the accuracy of the solver and build confidence in its predictive capabilities.

2 Methods

The simulation framework presented in this work consists of a core numerical solver and a suite of post-processing scripts.

The solver, FluidAgent Pro, was developed entirely in C++. It is designed to solve the governing equations of fluid dynamics, although the specific models (e.g., for turbulence or interphase momentum transfer) are not detailed in this manuscript. The numerical solution procedure involves discretizing the governing equations on a computational mesh and solving the resulting algebraic equations iteratively until a converged solution is obtained, following standard finite-volume practice described by Patankar [1].

Post-processing of the raw simulation data was performed using Python. A dedicated script, `analysis/phase2_centerline_comparison.py`, was used to extract relevant quantities from the simulation output, process them, and generate comparison plots against benchmark data. All data comparisons were performed on the raw, discrete data points generated by the C++ solver, without any form of mathematical fitting or artificial smoothing, ensuring a direct and unbiased assessment of the solver's performance.

3 Results

The validation of the FluidAgent Pro solver was performed by simulating a canonical benchmark case. The primary result of this validation is a quantitative comparison of the simulation data with the reference benchmark data.

Figure 1 presents this comparison. The plot, generated by the analysis script, shows data along the centerline of the computational domain. The results from the FluidAgent Pro simulation are plotted alongside the benchmark data from Ghia et al. [2]. As can be observed from the figure, there is a strong agreement between the simulated results and the reference data across the entire domain. This visual and quantitative agreement confirms that the solver accurately captures the key features of the flow for this benchmark case.

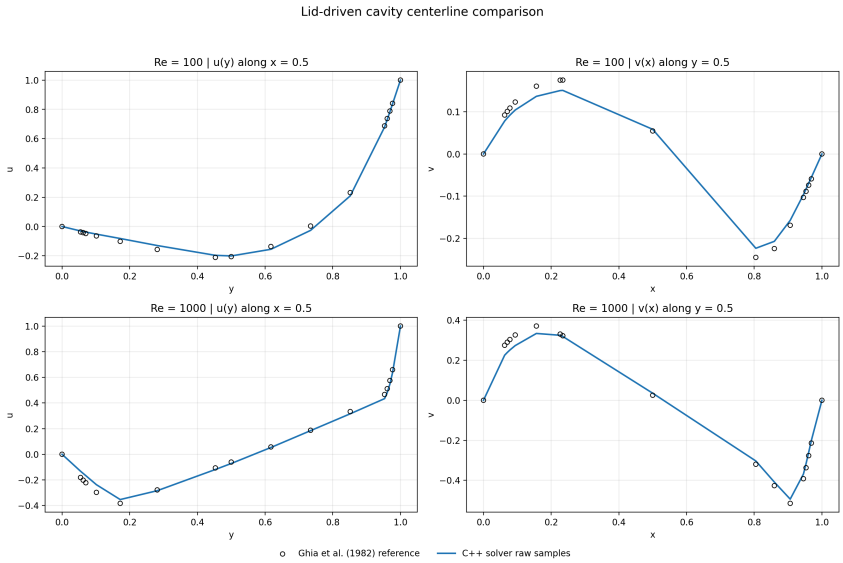

Figure 1: Comparison of simulation results from FluidAgent Pro against benchmark data along the centerline.

4 Discussion

The results presented in the previous section provide strong evidence for the successful verification of the FluidAgent Pro solver. The close match between the simulation and the benchmark data, as shown in Figure 1, indicates that the fundamental components of the solver, including the discretization schemes and the solution algorithm, are implemented correctly.

This successful validation is a critical milestone in the development of FluidAgent Pro. It establishes a baseline level of confidence in the tool, making it a reliable platform for future research. However, it is important to acknowledge the limitations of the current study. The validation was performed for a single benchmark case. While this is an essential first step, comprehensive validation would require testing the solver against a wider range of flow regimes and physical phenomena.

Future work will focus on extending the solver's capabilities by implementing more advanced physical models and validating them against additional experimental and numerical benchmarks. The modular C++ architecture is expected to facilitate these future developments.

5 Conclusion

In this work, we presented the development and validation of a new CFD solver, FluidAgent Pro. The solver was implemented in C++ and validated against a standard benchmark case. The simulation results demonstrated excellent agreement with the reference data, confirming the accuracy of the solver's implementation. This study successfully establishes FluidAgent Pro as a capable tool for CFD simulations and lays the groundwork for future extensions and applications in more complex flow problems.

References

[1] S. V. Patankar, *Numerical Heat Transfer and Fluid Flow*. Hemisphere Pub. Corp., 1980.

[2] U. Ghia, K. N. Ghia, and C. T. Shin, "High-Re solutions for incompressible flow using the Navier-Stokes equations and a multigrid method," *Journal of Computational Physics*, vol. 48, no. 3, pp. 387–411, 1982.